Assignment-II

Dataset

Windspeed (M/s)	Blade Angle (°)	Rotor Speed (RPM)	Power Output (KW)
5	10	120	45
7	12	150	70
9	14	180	102
11	16	210	138
13	18	240	180

$$X = \begin{bmatrix} 1 & 5 & 10 & 120 \\ 1 & 7 & 12 & 150 \\ 1 & 9 & 14 & 180 \\ 1 & 11 & 16 & 210 \\ 1 & 13 & 18 & 240 \end{bmatrix} \qquad Y = \begin{bmatrix} 45 \\ 70 \\ 102 \\ 138 \\ 180 \end{bmatrix}$$

$$X^T = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 5 & 7 & 9 & 11 & 13 \\ 10 & 12 & 14 & 16 & 18 \\ 120 & 150 & 180 & 210 & 240 \end{bmatrix} \Rightarrow X^T X = \begin{bmatrix} 5 & 45 & 70 & 900 \\ 45 & 445 & 670 & 9150 \\ 70 & 670 & 1020 & 13500 \\ 900 & 9150 & 13500 & 167400 \end{bmatrix}$$

$$X^T Y = \begin{bmatrix} 535 \\ 5671 \\ 8346 \\ 105840 \end{bmatrix}$$

$$\left[\theta = (X^T X)^{-1} X^T Y\right]$$ ie,

$\therefore$ $\theta = \begin{bmatrix} \cancel{-11.73} \\ \cancel{4.43} \\ \cancel{2.10} \\ \cancel{0.18} \end{bmatrix}$ $\theta = \begin{bmatrix} -11.5 \\ 6.75 \\ 4.25 \\ 0.18 \end{bmatrix}$

$\therefore$ ML model becomes.

$$Y = -11.5 + 6.75\, x_1 + 4.25\, x_2 + 0.18\, x_3$$

here,

$x_1 \rightarrow$ Wind speed.

$x_2 \rightarrow$ Blade angle.

$x_3 \rightarrow$ Rotor speed.

$\therefore$ When

$x_1 = 9$

$x_2 = 11$

$x_3 = 100$

$$Y = -11.5 + 6.75(9) + 4.25(11) + 0.18(100)$$

$$Y = -11.5 + 60.75 + 46.75 + 18$$

$$Y = 114 \text{ KW}$$